Computational HW 1

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Radioactive Decay

1 Introduction

Consider a radioactive decay problem involving two types of nuclei, A and B, with populations $N_A(t)$ and $N_B(t)$. Suppose that type A nuclei decay to form type B nuclei, which then also decay, according to the differential equations

$$\frac{dN_A}{dt} = -\frac{N_A}{\tau_A} \tag{1}$$

$$\frac{dN_B}{dt} = \frac{N_A}{\tau_A} - \frac{N_B}{\tau_B} \tag{2}$$

Here, we assume that type A nuclei are naturally occurring, and type B are not. This paper focuses on determing the age of an object by measuring the amount of type A and B nuclei within the object. We then discuss the accuracy of the age based on the ratio of measured nuclei N_B/N_A

2 Methods

The original Euler method is used. The form

$$f(t + \Delta t) = f(t) + \frac{df}{dt}(t)\Delta t$$

is adapted into our coupled differential equations. We obtain the iteration relations

$$N_A(t + \Delta t) = N_A(t) - \frac{N_A(t)}{\tau_A}\Delta t \tag{3}$$

$$N_B(t + \Delta t) = N_B(t) + \left(\frac{N_A(t)}{\tau_A} - \frac{N_B(t)}{\tau_B}\right)\Delta t \tag{4}$$

The following parameters were initialized as such:

1. Initial Type A nuclei $N_A(0) = 1$ *(Note: this represents 100% of the starting nuclei being type A.)*
2. Initial Type B nuclei $N_B(0) = 0$
3. Type A Decay Constant τ_A
4. Type B Decay Constant τ_B
5. Nuclei ratio $r = N_B/N_A$
6. Scaled time $T = t/\tau_A$

Algorithm 1 Euler's Method

Require: $t > 0$
  $N_A \leftarrow [1]$ ▷ Initial Value
  $N_B \leftarrow [0]$ ▷ Initial Value
  while $t < t_{MAX}$ **do**
    append $N_A(t) - \frac{N_A(t)}{\tau_A}\Delta t$ to N_A
    append $N_B(t) + \left(\frac{N_A(t)}{\tau_A} - \frac{N_B(t)}{\tau_B}\right)\Delta t$
    $t \leftarrow t + \Delta t$
  end while
  $r \leftarrow N_B/N_A$ ▷ Element-Wise Division
  return r

The analytic solutions for the coupled differential equations (1),(2) are:

$$N_A(t) = e^{-t/\tau_A} \tag{5}$$

$$N_B(t) = \frac{\tau_B}{\tau_A - \tau_B}\left(e^{-t/\tau_A} - e^{-t/\tau_B}\right) \tag{6}$$

The above algorithm was used, in conjunction with the analytic solution (see Appendix A.) to test the accuracy of the iteration method and explore uncertainty in age of the observed object. No use of generative AI was used in this assignment.

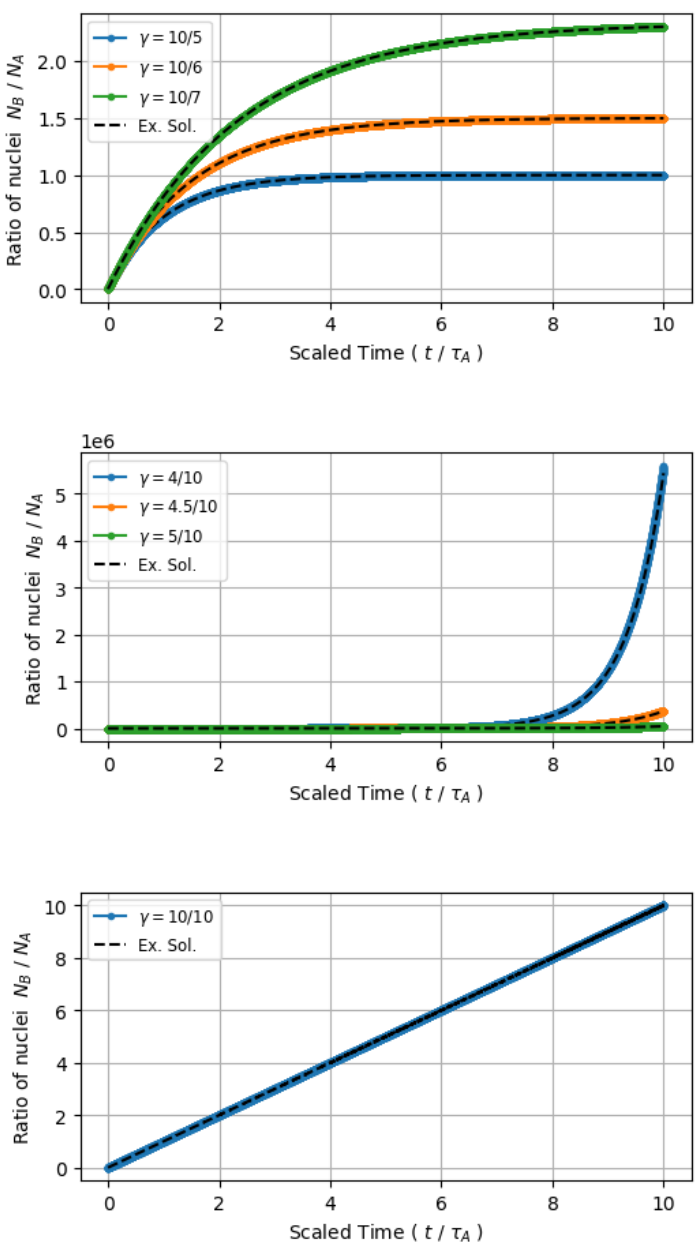

Figure 1: Ratio of nuclei types over time for different γ values (a) $\gamma > 1$, (b) $\gamma < 1$, (c) $\gamma = 1$

3 Results

Different decay ratios $\gamma = \tau_A/\tau/B$ were tested. Exponential bounded growth was observed with $\gamma > 1$, unbounded growth with $\gamma < 1$, and linear growth with $\gamma = 1$.

The ratios r and analytic solutions are plotted against corresponding scaled times T, shown in Figure 1.

4 Discussion

Euler's method proved consistent and in agreement with the analytic solution when for all $\gamma > 0$.

With $\gamma \geq 1$, shown in Figures (a) and (c), the age of the object can be accurately predicted for low scaled time T. Euler's method proved reliable for approximating the behavior of nuclei decay, with fractional differences less than 1%, decreasing as $\gamma \to \infty$.

However, by the behavior in Figure 1(a) shows unreliable age prediction as age increases. The bounded exponential growth shows little change in the ratio N_B/N_A despite large changes in scaled time. The predicted age cannot be precisely picked out among nearly identical results.

Shown in Figure 1.b, the age can be determined somewhat accurately at early ages. Though the exponential growth explodes, Euler's method is in agreement with the exact solution at all times.

On the other hand, a ratio of exactly $\gamma = 1$ showed consistently acceptable predictions. With a maximum error of 1%, and perfectly linear behavior, every value of scaled time T has a unique, corresponding ratio. Neither the exact solution nor the approximation out-grow the other.

It was found that large time steps Δt greatly increased the fractional difference between predicted and observed results. For example, the exponential growth shown when $\gamma < 1$ had large dissagreement when $\Delta t = 0.1$, with average error being $17\% - 32\%$. However, reducing the time step to $\Delta t = 0.01$ reduced the error range down to $1.6\% - 2.2\%$. With proper precautions, Euler's method can be reliably used to follow the analytic solution of this decay pattern.

5 Summary

Euler's Method proved to be very reliable in predicting the age of an object based on the ratio of N_A and N_B type nuclei within it. The time constant ratio γ determined the decay behavior, which in turn provided varying results on age prediction accuracy.

Derivation of Analytical Solution

We start with the coupled differential equations:

$$\frac{dN_A}{dt} = -\frac{N_A}{\tau_A}$$

$$\frac{dN_B}{dt} = \frac{N_A}{\tau_A} - \frac{N_B}{\tau_B}$$

The first can be easily seen:

$$N_A'(t) = \frac{-1}{\tau_A} N_A(t) \rightarrow \boxed{N_A(t) = e^{-t/\tau_A}}$$

For our case, the initial value is simply 1.
This result is then used for N_B:

$$N_B'(t) = \frac{N_A(t)}{\tau_A} - \frac{N_B(t)}{\tau_B}$$

$$N_B'(t) + \frac{N_B(t)}{\tau_B} = \frac{1}{\tau_A} e^{-t/\tau_A}$$

We find the integrating factor:

$$e^{\int dt/\tau_B} = e^{t/\tau_B}$$

Multiply the ODE by this integrating factor:

$$e^{t/\tau_B} N_B'(t) + \frac{1}{\tau_B} E^{t/\tau_B} N_B(t) = \frac{1}{\tau_A} e^{t/\tau_B - t/\tau_A}$$

Recognize the chain rule:

$$\frac{d}{dt}\left[e^{t/\tau_B} N_B(t)\right] = \frac{1}{\tau_A} e^{t/\tau_B - t/\tau_A}$$

Integrate both sides:

$$\int \frac{d}{dt}\left[e^{t/\tau_B} N_B(t)\right] dt = \int \frac{1}{\tau_A} e^{t/\tau_B - t/\tau_A}$$

$$e^{t/\tau_B} B(t) = \frac{1/\tau_A}{1/\tau_B - 1/\tau_A}\left[e^{t/\tau_B - t/\tau_A} - 1\right]$$

$$\boxed{B(t) = \frac{\tau_B}{\tau_A - \tau_B}\left(e^{-t/\tau_A} - e^{-t/\tau_B}\right)}$$